MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 02** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer question (2 marks) and 3 Short Answer questions (3 marks each).
4. **Section C** comprises 1 Applied Case Study question of 4 marks with internal choice.
5. Use of calculators is not permitted.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. The first four terms of an AP whose first term is -2 and common difference is -2 are: [1]

(A) $-2, 0, 2, 4$ (B) $-2, 4, -8, 16$ (C) $-2, -4, -6, -8$ (D) $-2, -4, -8, -16$

Q2. The 21st term of the AP whose first two terms are -3 and 4 is: [1]

(A) 17 (B) 137 (C) 143 (D) -143

Q3. If the 2nd term of an AP is 13 and the 5th term is 25, what is its 7th term? [1]

(A) 30 (B) 33 (C) 37 (D) 38

Q4. Which term of the AP $21, 42, 63, 84, \ldots$ is 210? [1]

(A) 9th (B) 10th (C) 11th (D) 12th

Q5. **Assertion (A):** The nth term of the AP $a, a+d, a+2d, \ldots$ is given by $a_n = a + (n-1)d$.
Reason (R): Each term is obtained by adding the common difference d to the preceding term. [1]

(A) Both (A) and (R) are true; (R) explains (A). (B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Short Answer Questions (11 Marks)

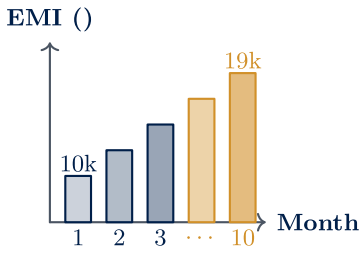
Q6. Which term of the AP $3, 15, 27, 39, \ldots$ will be 120 more than its 21st term? [2]

Q7. Find the sum of all two-digit natural numbers which leave a remainder 1 when divided by 4. [3]

Q8. The angles of a quadrilateral are in AP. If the greatest angle is double the least angle, find all four angles of the quadrilateral. [3]

Q9. Find three numbers in AP whose sum is 21 and the product of the first and third numbers exceeds the second number by 6. [3]

Q10. CASE STUDY: Automobile Bank Loan Repayment Amortization



Rishi purchased a brand-new electric SUV and financed it with an auto-loan from a nationalized bank. To encourage disciplined early repayment, the bank offers a stepped repayment scheme.
Rishi agrees to pay his total loan through monthly installments. He starts with a first installment of 10,000 and commits to increasing each subsequent monthly installment by 1,000 over the preceding month until the entire loan is completely discharged.

Based on the above mathematical scenario, answer the following questions:

(i) Write down the first three terms of the AP representing the monthly installments paid by Rishi. [1]

(ii) What amount does Rishi pay to the bank in his 10th monthly installment? [1]

(iii) What total cumulative amount has Rishi paid to the bank in the first 12 installments (1 full year)? [2]

— OR —

If the total loan principal amount to be cleared was 1,18,000, how many monthly installments will he need to clear the debt? [2]

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